

Basic Principles of Medicine II
Module: Nervous System & Behavioral
Science

Synaptic Transmission

Lecture No: NB 08

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Recommended Reading

- Synaptic Transmission | Essential Neuroscience, 4e | Medical Education | Health Library (oclc.org)

Lecture Objectives

- | | |
|------------------------------------|--|
| SOM.MK.I.BPM2.1.NB.1.
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| SOM.MK.I.BPM2.1.NB.1.
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| SOM.MK.I.BPM2.1.NB.1.
NSCI.0094 | List five steps in the life cycles of small-molecule transmitters |
| SOM.MK.I.BPM2.1.NB.1.
NSCI.0095 | List five steps in the life cycles of neuroactive peptides |
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| SOM.MK.I.BPM2.1.NB.1.
NSCI.0100 | Explain mechanisms for destruction/removal of neurotransmitter from the synapse |
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Electrical Synapses

Synchronize electrical activity among cells

Properties:

- Mediated by gap junctions
- Voltage-sensitive
- Pass ions directly between cells
- Rapid intercellular communication rapid

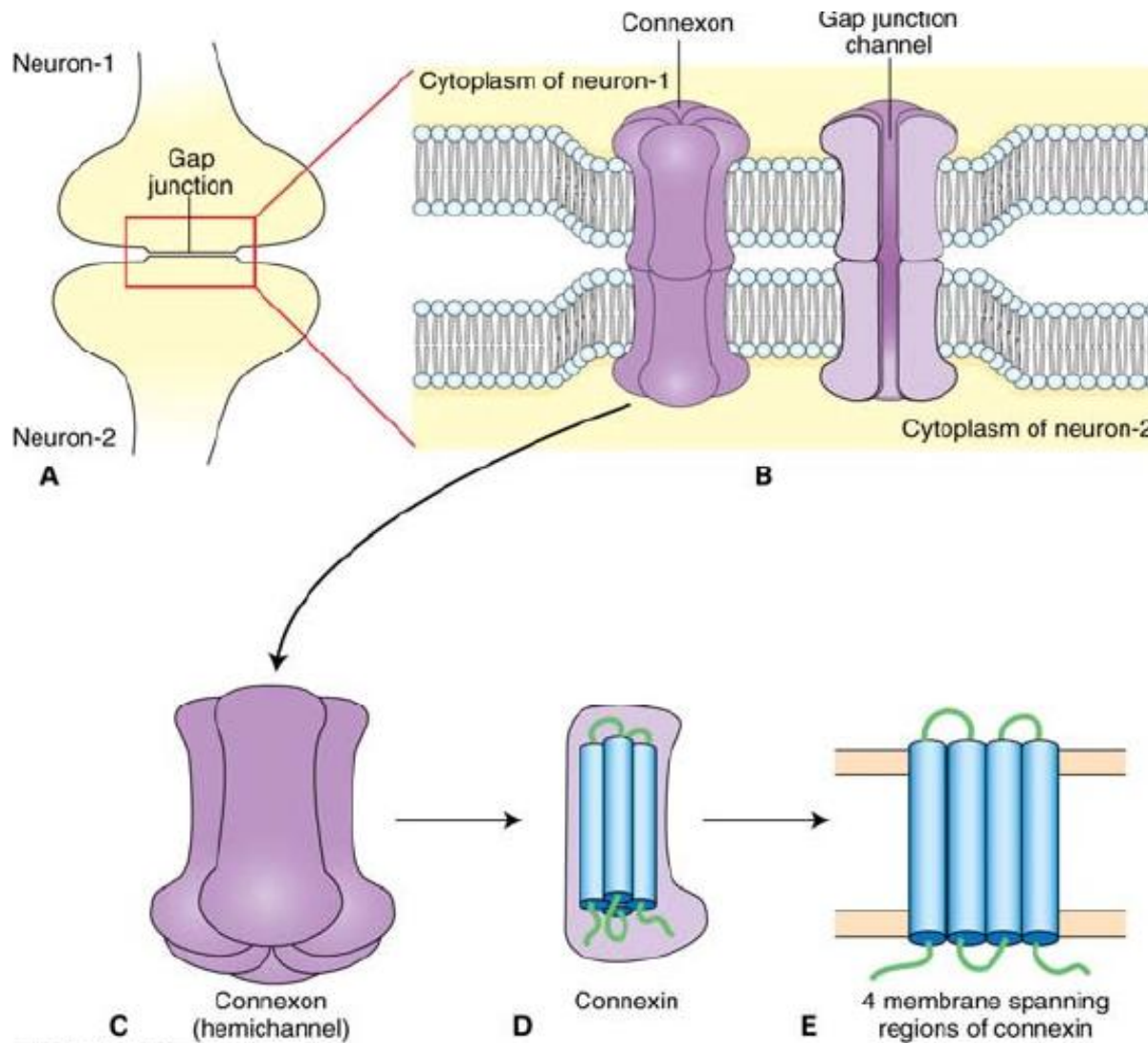
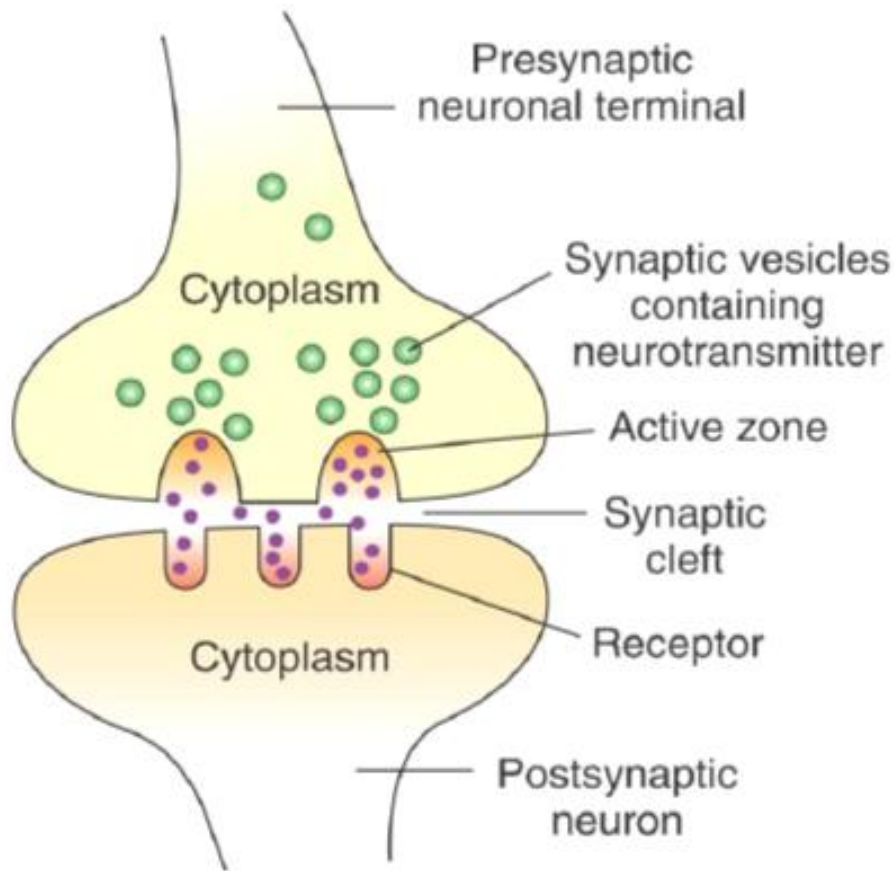


Figure 6.1, page 96, in 'Essential Neuroscience' -by Allan Siegel and Hreday N. Sapru, Lippincott Williams & Wilkins, 2015.

Chemical Synapses



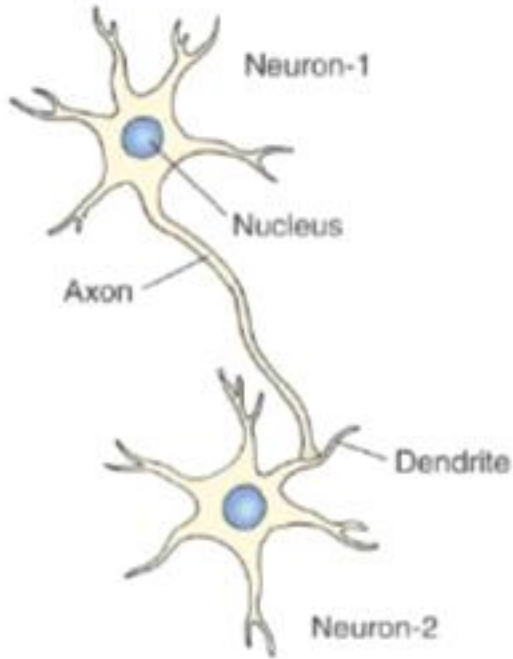
A Chemical synapse

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*Figure 6.2, in 'Essential Neuroscience' - by
Allan Siegel and Hriday N. Sapru, Lippincott
Williams & Wilkins, 2015.*

- Most common synaptic contact among mammalian neurons
- Presynaptic cells release chemicals that bind to receptors
 - Post-synaptic receptors mediate altered postsynaptic neuronal function
 - Presynaptic receptors regulate exocytosis
- Synapses can persistently change in function to mediate learning

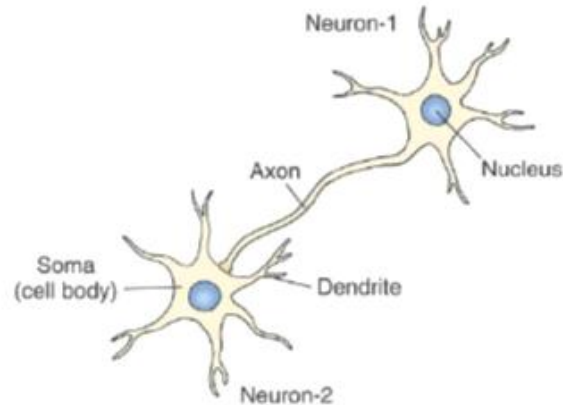
Chemical Synapses: Location and Function



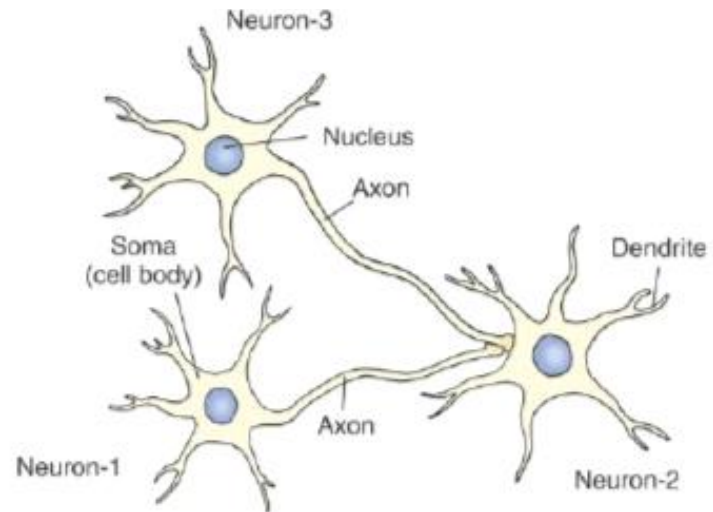
Axo-dendritic synapse

Influence likelihood of AP at target cell:

Distal synapses tend to be excitatory; proximal synapses tend to be inhibitory

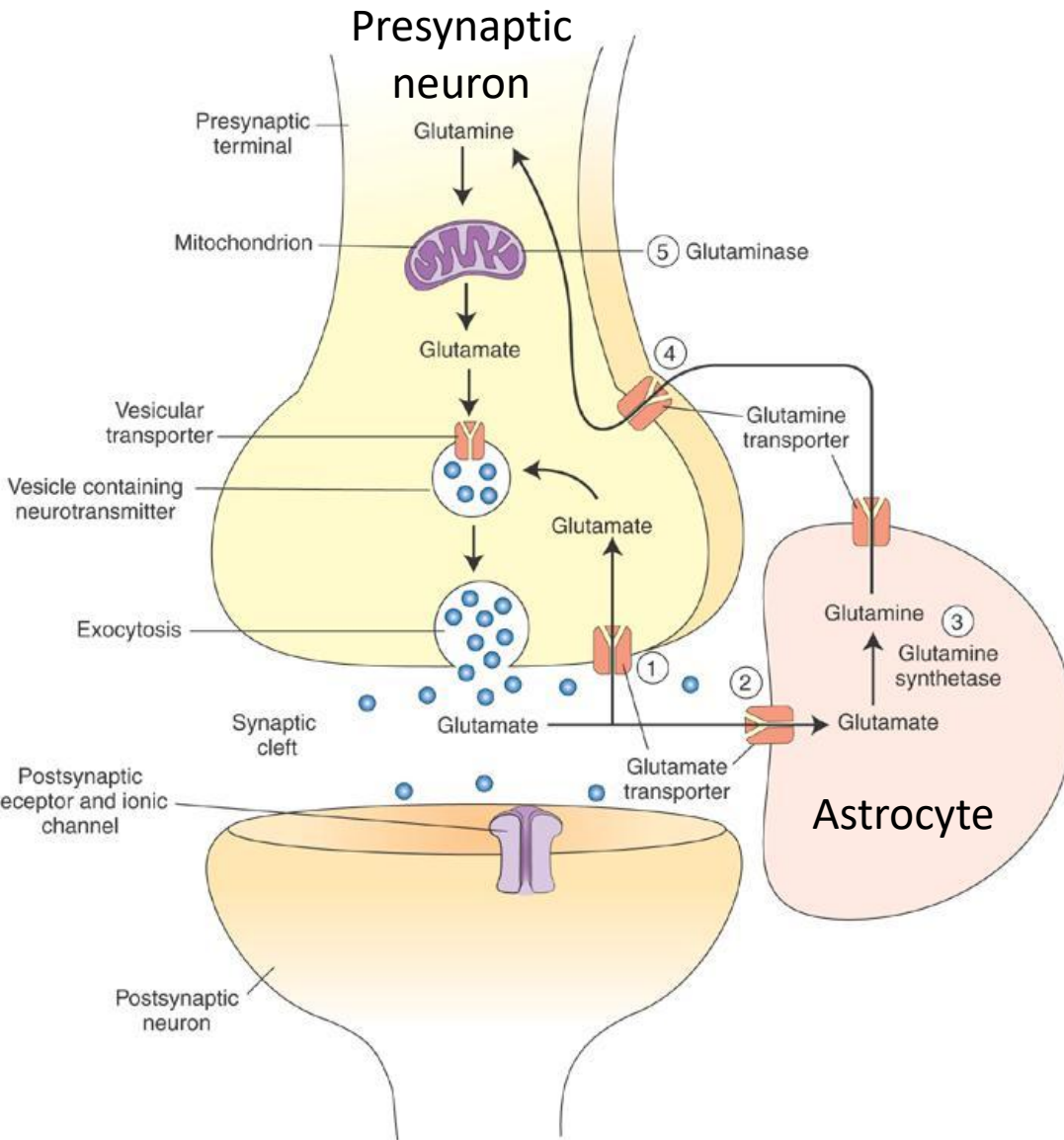


Axosomatic Synapse: Often reduce probability of AP at target cell



Axoaxonic Synapse: Diminish magnitudes of APs and thus reduce transmitter release

Tripartite Synapses

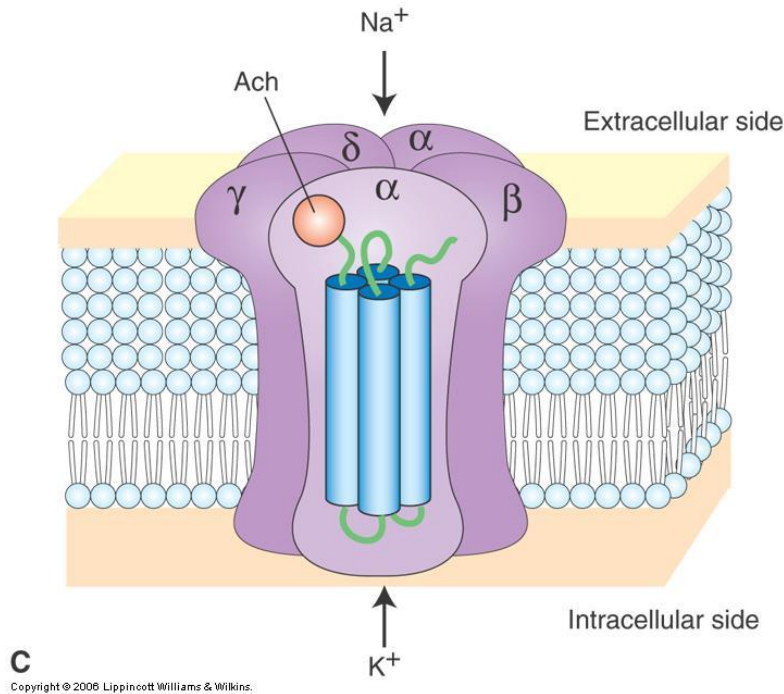


- Astrocytes take up glutamate released by neurons
- Glutamate metabolized to glutamine
- Glutamine returned to neurons for conversion back to glutamate

Lecture Objectives

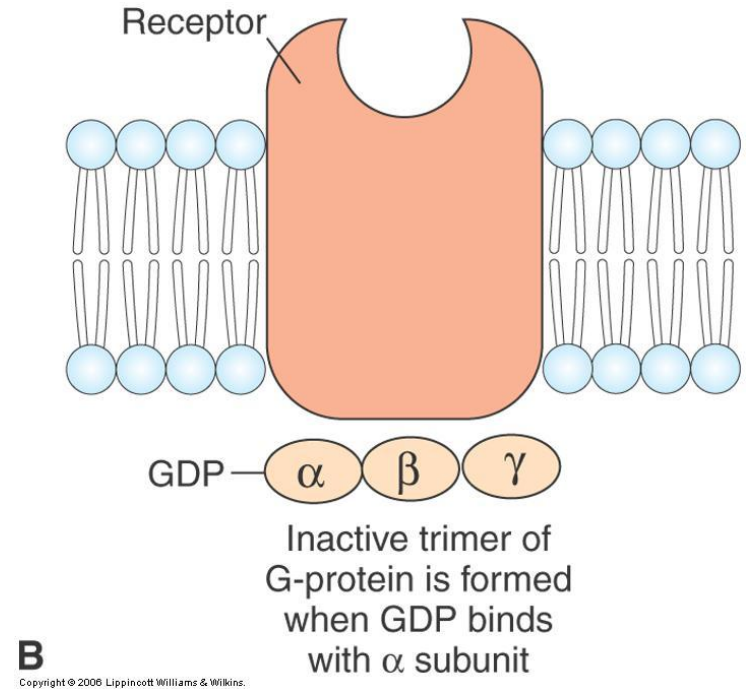
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ACh Receptors: Ionotropic vs Metabotropic



Ionotropic: Nicotinic

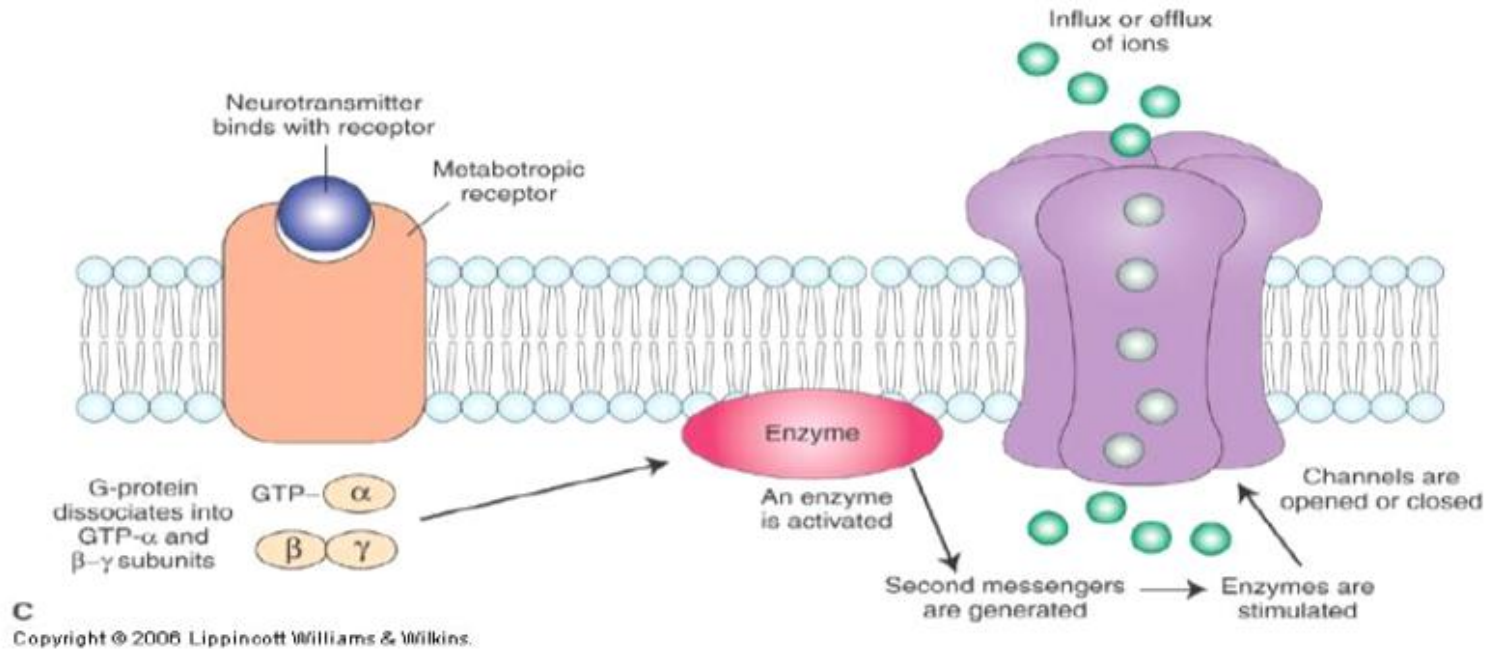
Change in ion flux (Na^+ , K^+)
Rapid effects (milliseconds)



Metabotropic: Muscarinic

Metabolic changes (e.g., phosphorylation)
Slower effects (seconds to minutes)

Metabotropic receptors

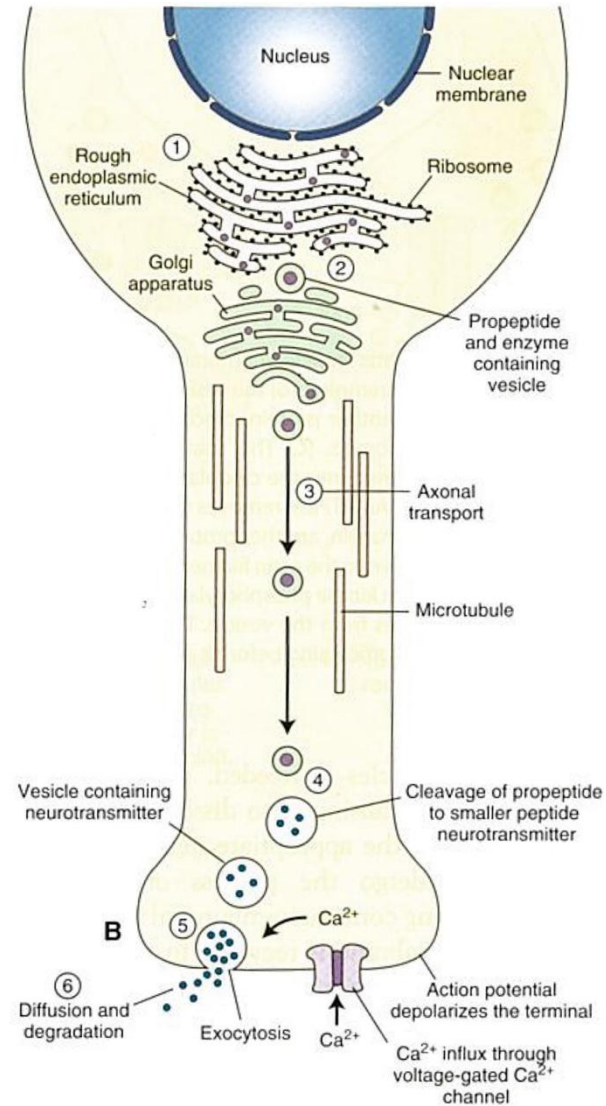
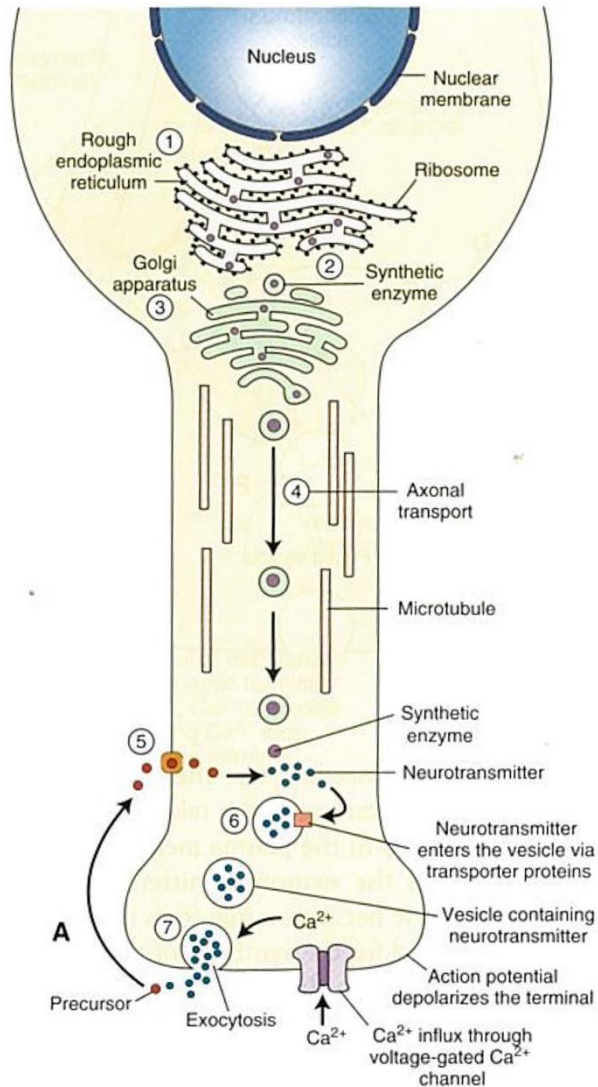


- Receptor coupled to G-proteins
- Ligand binding dissociates α subunit from $\beta\gamma$ complex
- α subunit can be stimulatory or inhibitory
 - Effects on ion channels, metabolism and gene expression

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Life Cycles of Transmitters



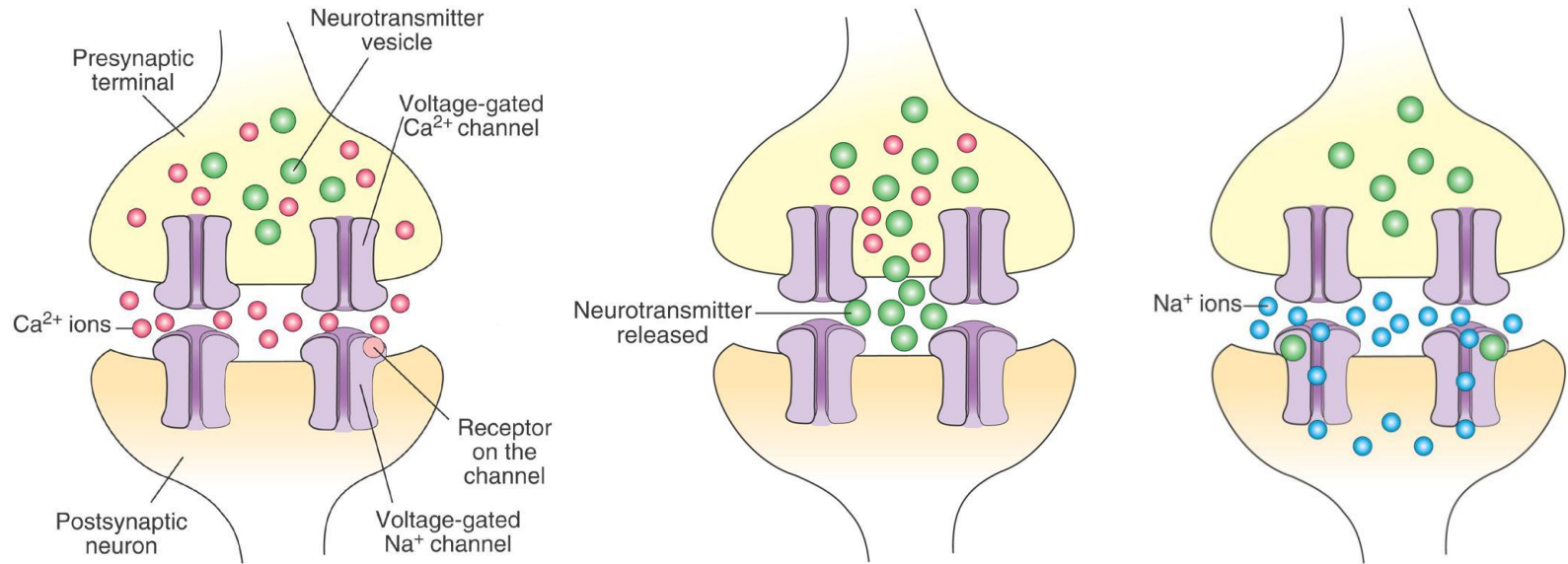
Life Cycles of Transmitters

Feature	Small-molecule neurotransmitters	Large-molecule neurotransmitters (Neuropeptides)
Examples	Glutamate, GABA, glycine, acetylcholine, dopamine, serotonin, norepinephrine	Substance P, endorphins, enkephalins, neuropeptide Y, somatostatin
Site of synthesis	Presynaptic terminal (enzymes synthesized in soma, transported to terminal)	Cell body (soma)
Precursor source	Amino acids or simple metabolites taken up by terminal	mRNA → pre-propeptides synthesized on RER
Processing	One- or few-step enzymatic reactions in terminal	Extensive post-translational processing (cleavage, modification) in ER/Golgi
Vesicle type	Small, clear-core vesicles	Large, dense-core vesicles
Speed of action	Fast (milliseconds)	Slow onset, long-lasting
Termination of action	Reuptake into nerve terminal or glia; enzymatic degradation in cleft	Diffusion away; enzymatic degradation by extracellular peptidases
Recycling	Transmitter, metabolite, vesicles recycled	No recycling

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Synaptic Delay: 0.5 -100 MS



A Action potential depolarizes the presynaptic terminal membrane, Ca^{2+} channels open and Ca^{2+} ions enter the terminal.

B Ca^{2+} ions in the terminal promote fusion of vesicular and terminal membranes and the neurotransmitter is released in the synaptic cleft.

C Neurotransmitter binds with the receptor on the channel which opens, Na^{+} ions enter the postsynaptic neuron, depolarize it, and excite it.

Arrival of AP



1. Activation of voltage-gated Ca^{2+} channel
2. Fusion of vesicle and release of transmitter into cleft
3. Diffusion of transmitter across cleft to postsynaptic neuron
4. Transmitter-receptor binding
5. Ionic flux or induction of enzyme activity



Postsynaptic Response

Lecture Objectives

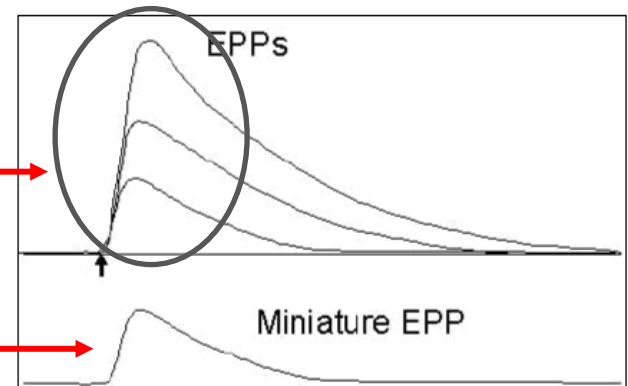
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Quantal Size

- Smallest post-synaptic response (relates to transmitter released from a single vesicle)
 - Miniature endplate potentials (i.e., minis)
 - Endplate potentials increase in size in relation to the number of coincident minis
- Demonstrated by preventing calcium from entering cell
 - Traces of intracellular calcium provoke random release from isolated individual vesicles

Postsynaptic reactions to transmitter from 1, 2 or 3 vesicles

Postsynaptic reaction to transmitter from a single vesicle



Modulation of Quantal Size

- Presynaptic
 - Varying numbers of transmitter molecules in vesicles
- Postsynaptic
 - Changing availability or sensitivity of receptors
 - Blocking receptors
 - Adding or destroying receptors
 - Sensitizing/desensitizing receptors
- Extracellular
 - Altering synaptic longevity of transmitter
 - Affecting degradation

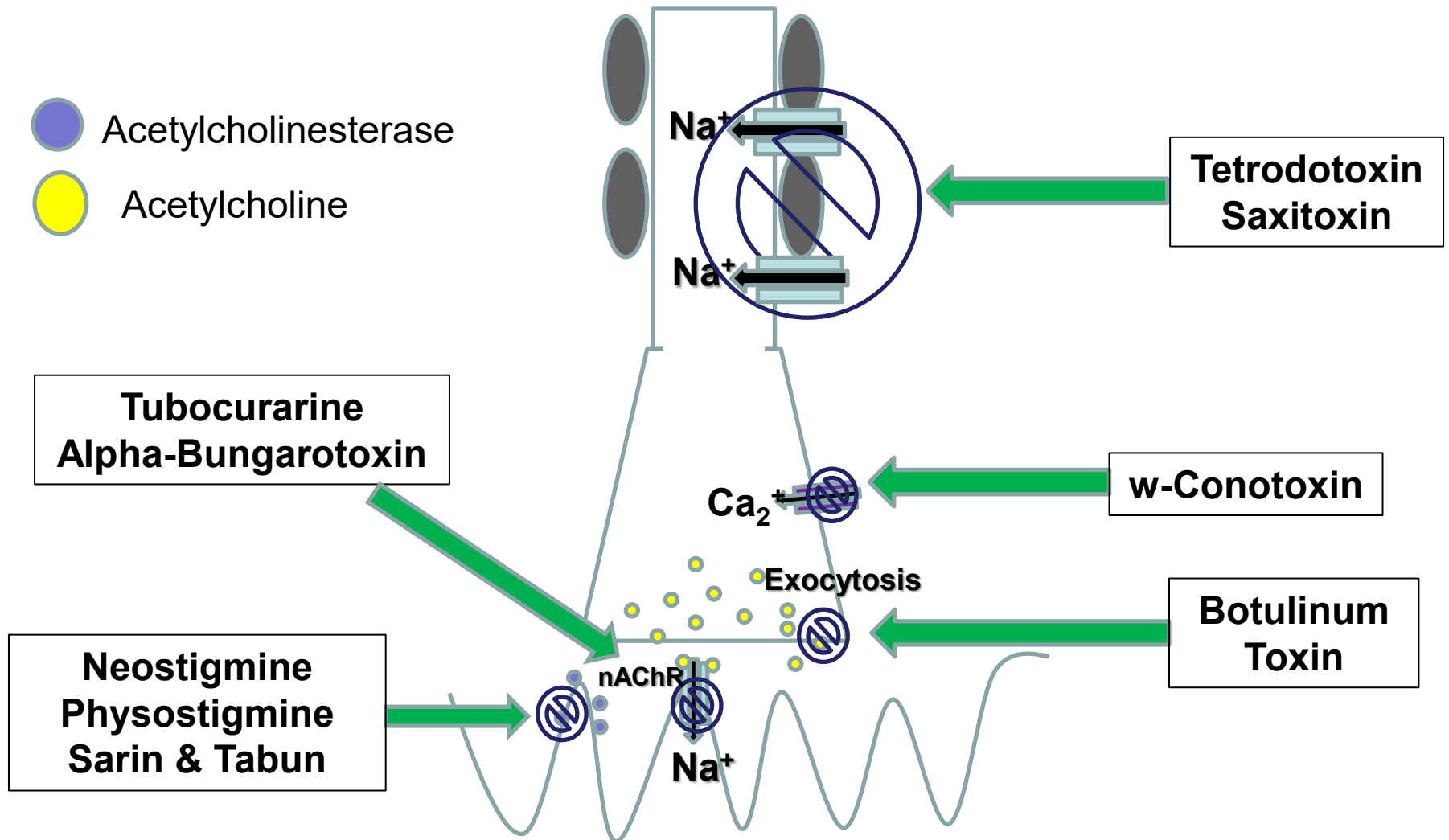
Quantal Content

- Number of vesicles released per action potential
- Modulation
 - Presynaptic
 - Adjusting calcium influx
 - Modulating action potentials
 - E.g., blocking potassium channels
 - Blocking/destroying calcium channels
 - Varying extracellular calcium concentrations

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Chemically Altered Release 1: Modulating ACh Release



Chemically Altered Release 1

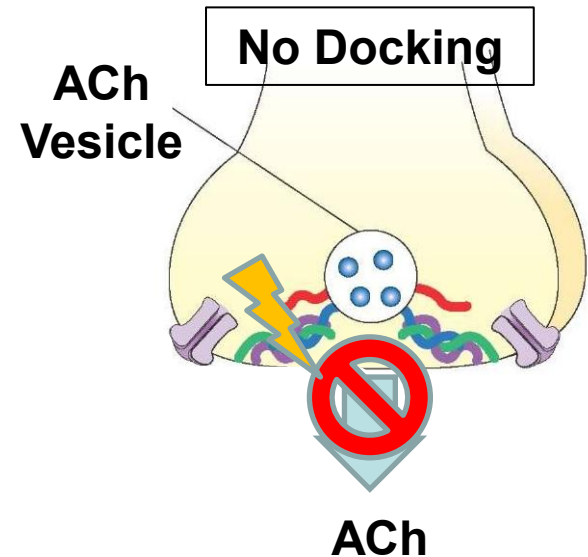
Modulating ACh Release

- Tetrodotoxin and saxitoxin
 - Block voltage-gated sodium channels
 - Blocks action potentials
- ω -Conotoxin
 - Blocks voltage-gated calcium channels
 - Reduces quantal content
- Botulinum toxin
 - Cleaves synaptobrevin
 - Reduces quantal content
- Neostygmine, physostigmine, Sarin, Tabin
 - Block acetylcholine esterase
 - Increases quantal size
- Tubocurarine, α -bungarotoxin
 - Block nicotinic receptors
 - Reduce quantal size

Chemically Altered Release 2:

Botulinum Toxin

- Protease produced by bacterium *Clostridium botulinum*
- Clinical symptoms
 - Diplopia (double-vision)
 - Dysphagia (difficulty swallowing)
 - Xerostomia (dry mouth)
 - Dysarthria (difficulty speaking)
 - Muscle weakness – limbs and trunk
 - Respiratory paralysis
- Therapeutic uses
 - Treatment of dystonia
- Cosmetic uses
 - Botox



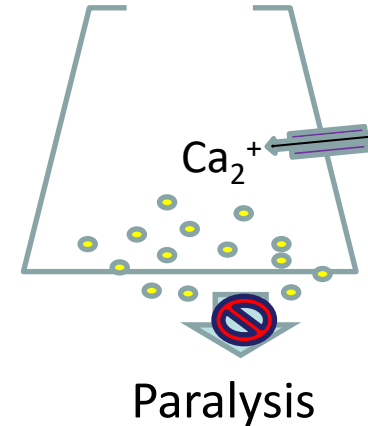
Protease breaks down synaptobrevin, therefore no vesicle docking and no ACh release. Irreversible!

Chemically Altered Release 3:

Tetanus Toxin

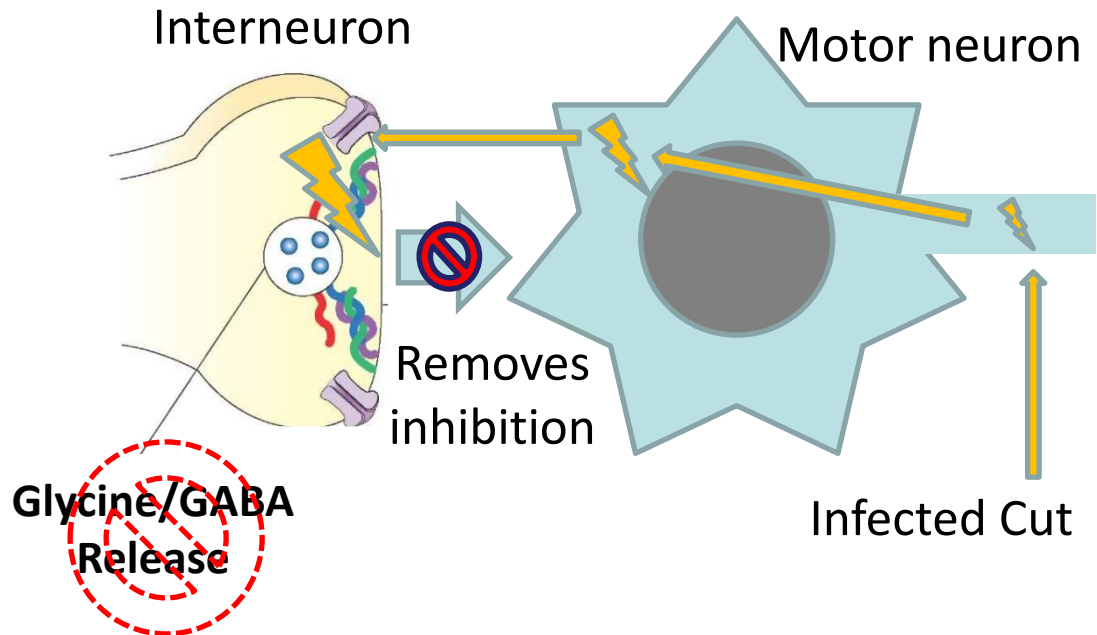
Beta-bungarotoxin

- Snake venom
- Binds to actin cytoskeleton
- Inhibits phosphorylation of synapsin 1
- No vesicle release



Tetanus Toxin

- *Clostridium tetani*
- Degrades synaptobrevin
- Inhibits glycine and GABA release
- Sustained powerful contractions (hypertonia)
- Irreversible

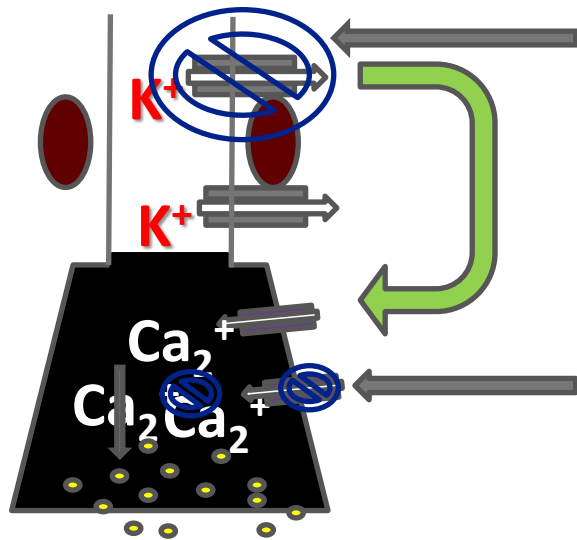


Chemically Altered Release 4:

Drugs Affecting Quantal Dynamics

Aminoglycoside antibiotics (Neomycin, Streptomycin)

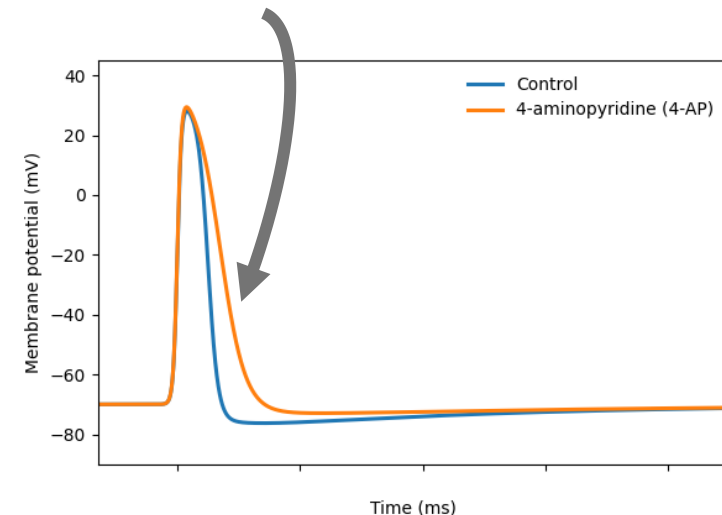
- Block presynaptic calcium channels (reversible with enhanced calcium influx)



4AP restores quantal content in lower motor neurons

4-Aminopyridine (K^+ channel blocker) increases duration of action potential, enhancing calcium influx

Aminoglycoside antibiotics block calcium channels, reducing quantal content (causing weakness)

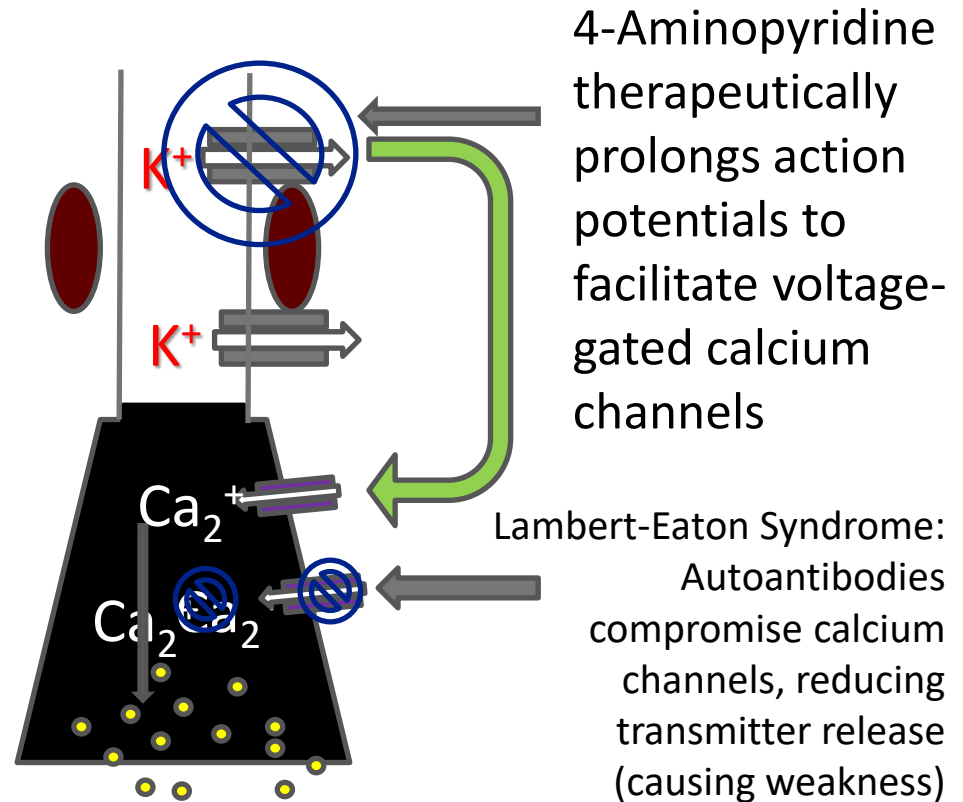
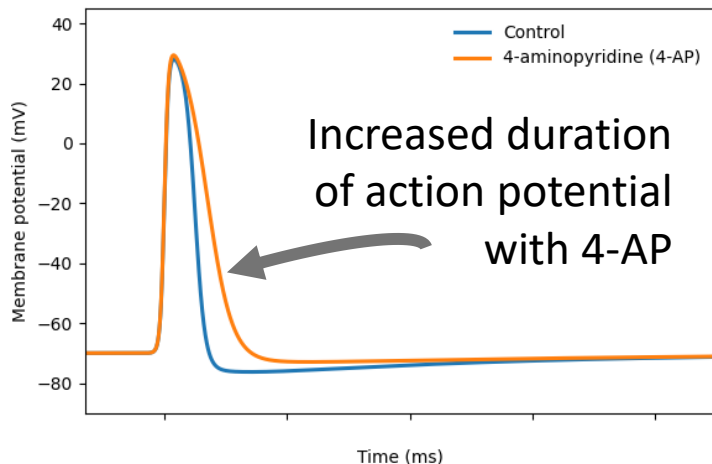


Chemically Altered Release 5

4-Aminopyridine Therapy

Lambert-Eaton Syndrome

- Risk factors
 - Cancer, typically small-cell carcinoma of the lung (SCCL)
- Causes muscle weakness if left untreated
- Sustained or repeated effort briefly increases strength



4AP restores quantal content in lower motor neurons in Lambert-Eaton syndrome by increasing the duration of action potentials

Quantal Content Quick Study Guide

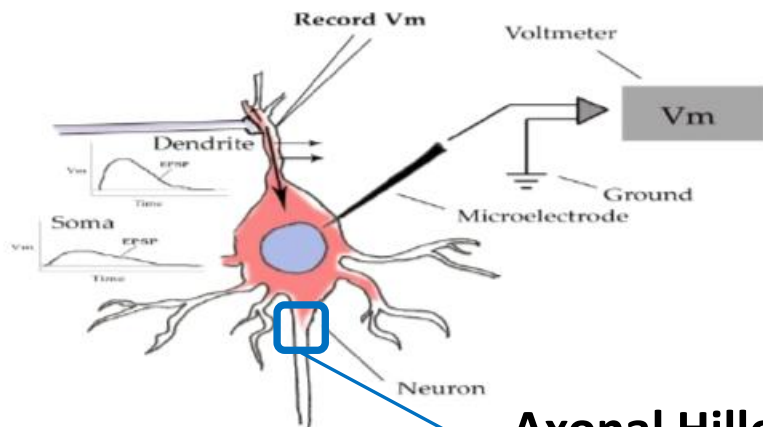
Target	Problem		Solution
	Antibiotics	Lambert-Eaton Syndrome	4-Aminopyridine
Voltage-gated Potassium Channels	Nil	Nil	Blockage
Voltage-gated Calcium Channels	Blockage	Blockage, Cross-linking, Internalization	Indirect Voltage-dependent Opening
Quantal Content	Reduced	Reduced	Increased
Muscle Strength	Reduced	Reduced	Increased

Lecture Objectives

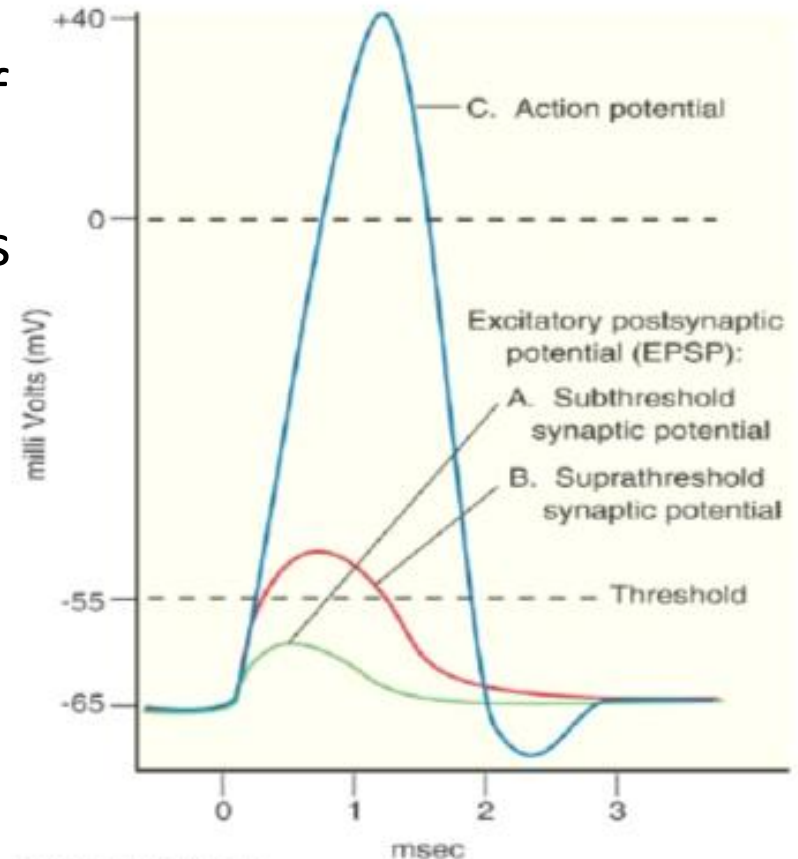
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Graded Potentials and EPSP

Neurotransmitter release produces sub-threshold graded potentials, brief local changes in dendrites and cell bodies but not axons. Current spreads with decrement according to length constant.



Axonal Hillock
Trigger Zone for APs
(greater density of NaV)



EPSP spatial and temporal summation

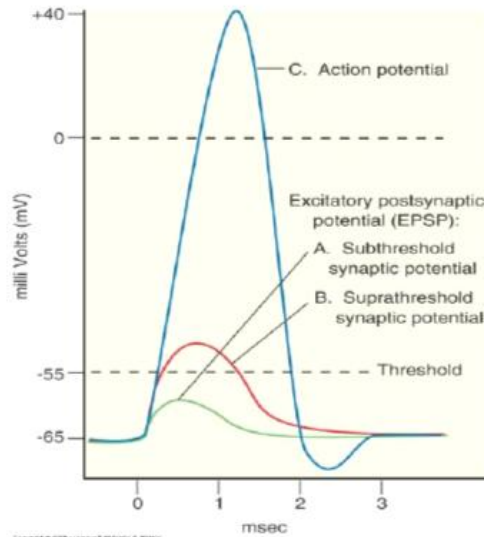
Spatial

Record EPSP

Temporal

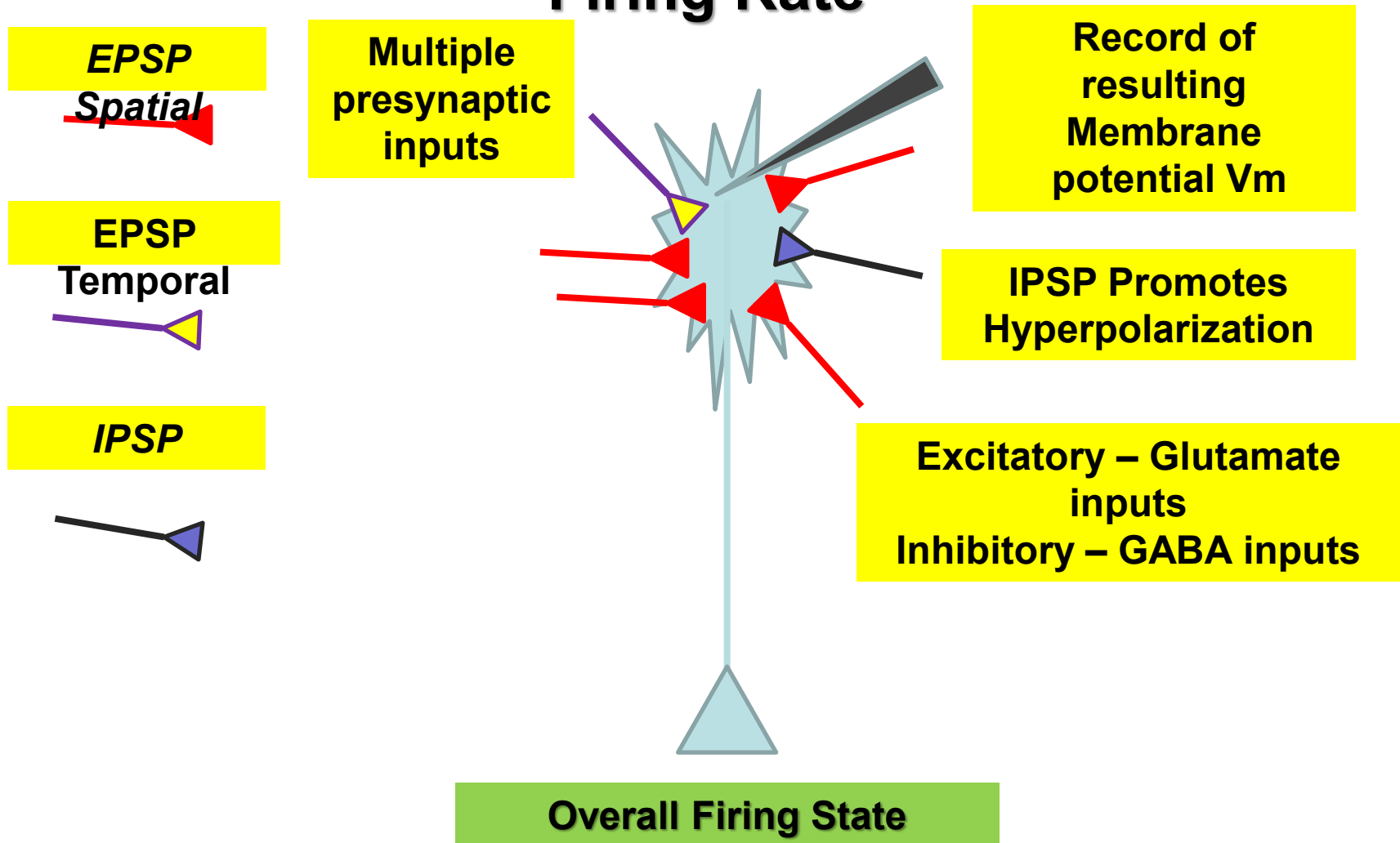
Multiple presynaptic inputs

Single presynaptic input – high frequency of impulses



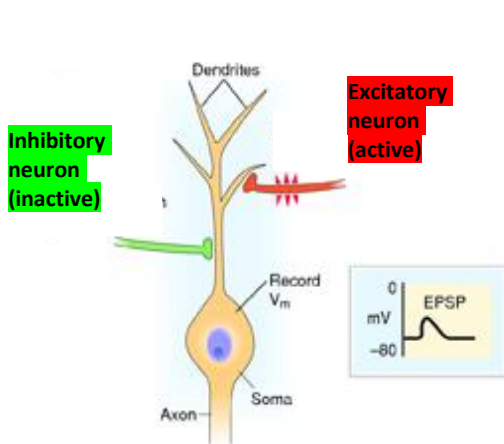
Summation = Multiple signals arriving at trigger zone are superimposed - additive effect

Summation: EPSP and IPSP Govern Overall Firing Rate

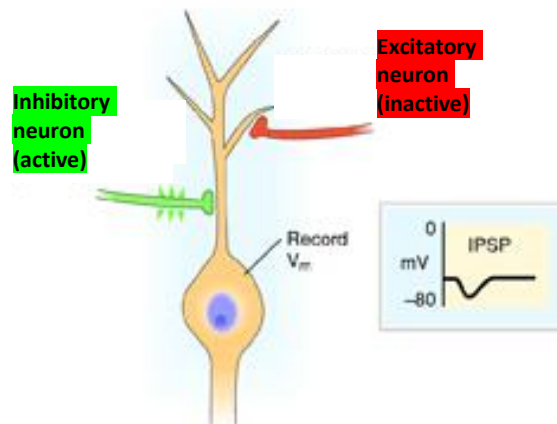


Neural signal integration

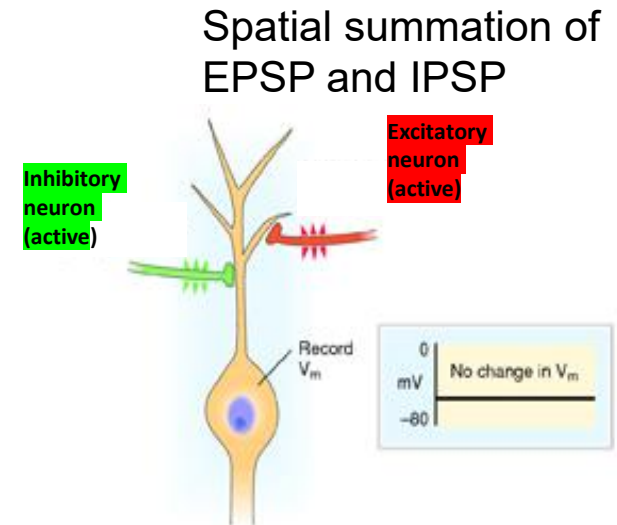
- Addition of all pre-synaptic neuronal inputs onto one neuron
- Can involve excitatory and inhibitory inputs (+ / -)
- Sum of all inputs determines if AP is generated



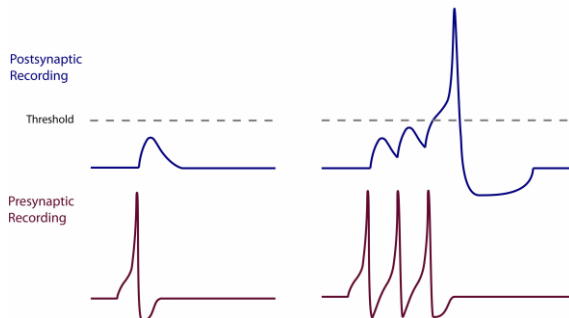
Excitatory input
-> firing of neuron
more likely



Inhibitory input
-> firing of neuron
less likely



Excitatory + Inhibitory input
-> cancel each other



Temporal summation of EPSPs

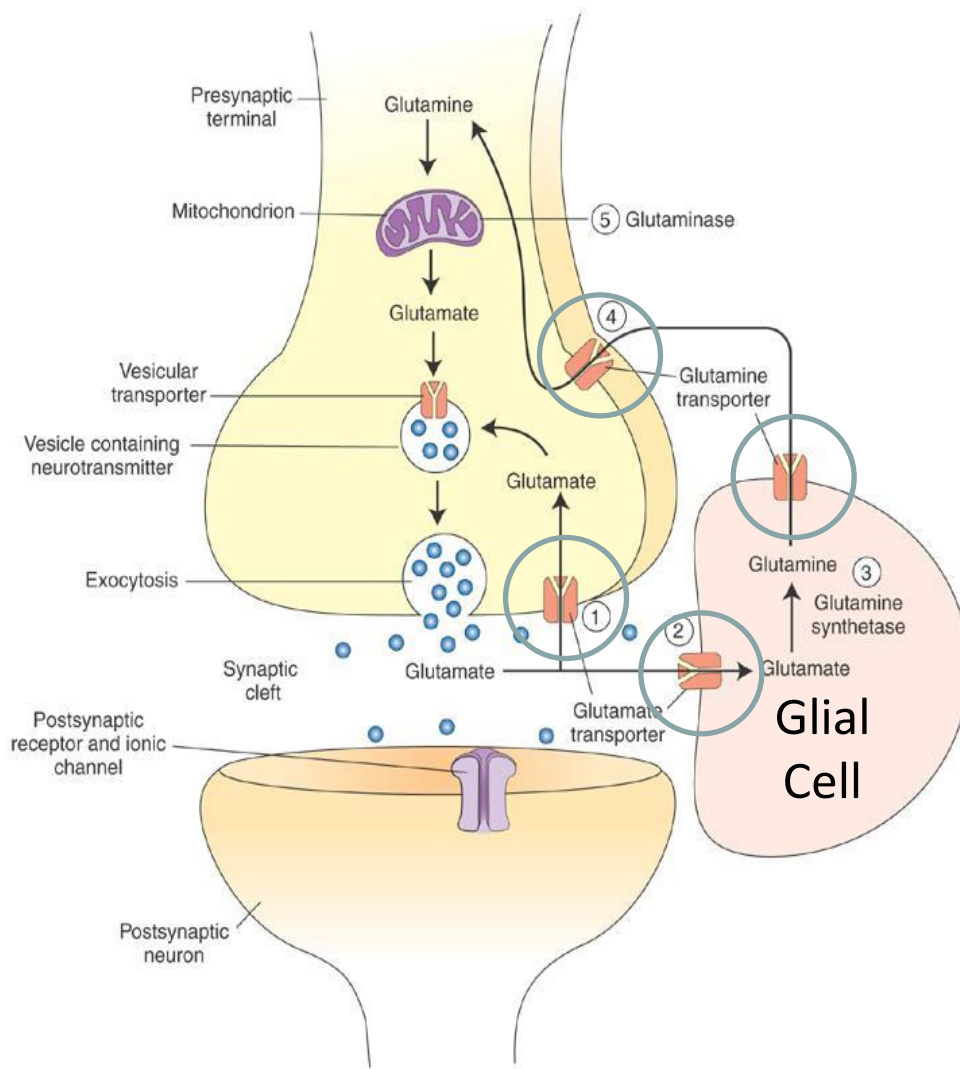
Temporal and spatial integration

- Two types
 - **Temporal summation**
 - Repeated high-frequency firing by a unitary presynaptic input -> changes likelihood of action potential firing
 - **Spatial summation**
 - Activity from different pre-synaptic neurons converge onto one neuron -> changes likelihood of action potential firing

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SOM.MK.I.BPM2.1.NB.1. NSCI.0099	Explain decrement and summation of postsynaptic potentials
SOM.MK.I.BPM2.1.NB.1. NSCI.0100	Explain mechanisms for destruction/removal of neurotransmitter from the synapse

Examples of NT removal/degradation



- Transporters remove most NTs from synaptic cleft
 - Except gases, ACh and peptides
- Monoamines (NE, EPI, DA, 5-HT) removed by pumps, then recycled or degraded
 - Degraded by MAO in pre-synaptic terminal, glia, liver
 - Degraded in dendrites, glia, liver by COMT (DA, NE, EPI)